

Mediating Digital Security: Participatory Learning with Older Adults in Smartphone Café Sessions

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Abstract

This study examines how older adults engage with digital technologies, focusing on privacy, security, and participation. Through community-based workshops, we explore usability challenges, fears around online risks, and the role of social learning. Our findings reveal how mediators help older adults navigate digital tools through trust-based support and peer learning. We argue that digital inclusion requires more than accessible design; it also depends on social infrastructures that enable confidence and shared problem-solving. This work contributes to participatory design (PD) by highlighting the value of mediated learning and community-driven support in fostering digital literacy among marginalized aging populations.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; • **Applied computing** → *Computers in other domains*; • **Social and professional topics** → **Digital divide**; • **Security and privacy** → *Social aspects of security and privacy*.

Keywords

Older Adult, Participatory Design, Digital Inclusion, Security and Privacy, Mediated Learning, Smartphone Café

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1 Introduction

The rapid proliferation of digital technologies has transformed how individuals communicate, access information, and conduct daily tasks. Tools such as smartphones, social media, online banking, and telehealth have become essential infrastructures [25, 50, 54]. However, this transformation has also deepened socio-technical disparities, particularly among older adults who often face complex barriers to digital engagement [48]. These barriers include age-related cognitive and physical changes and limited digital experience [7, 12, 37]. Although research in Human-Computer Interaction (HCI) and gerontechnology has addressed usability and social exclusion among older adults [12, 15, 16, 20], digital security remains a less developed area. Many older adults lack awareness of phishing, fraudulent apps, and complex privacy settings [21, 22, 56]. These challenges are even more pronounced for older migrants, who may also face language barriers, lower access to resources, and limited trust in digital systems.

Few studies have examined how informal community-based interventions could support security literacy and build sustained confidence in navigating digital environments. This paper investigates how participatory, trust-based learning environments can address these gaps. Drawing on six months of ethnographic fieldwork in

two small towns in western Germany, we explore the Smartphone Café initiative, an ongoing municipal program that supports older adults to build digital competency. Through peer-led sessions, personalized guidance, and informal social interactions, guided by trained mediators, these cafés foster an inclusive space where older adults can collaboratively navigate digital tools and security risks.

Our work contributes to PD discourse by conceptualizing these cafés as ‘participatory infrastructures’ [55], sustained, socially embedded environments where design practices are distributed across time, community, and context. Rather than treating older adults as one-time participants in design workshops, we examine how long-term mediation, social learning, and trust-building act as forms of everyday participation.

Research Questions. This study addresses the following questions:

1. How does mediation within participatory infrastructures shape older adults’ appropriation of smartphone practices for digital security literacy?
2. What cognitive, physical, and experiential barriers arise in everyday use?
3. Which mediating practices (material, social, discursive) enable confidence and safer behavior?
4. How do participants carry forward these practices beyond the Smartphone Café (peer help, routines)?

Contribution. We examine how older adults appropriate everyday smartphone practices in community Smartphone Cafés, focusing on digital security literacy (e.g., messaging, updates, payments) as it emerges through participatory infrastructures of mediation and peer support. This paper contributes to PD by concretizing participatory infrastructures through the everyday work of mediation in a municipal Smartphone Café. Empirically, we show how mediators enact material (devices, connectivity, scripts), social (trust, accompaniment), and discursive (risk framing, translation) roles over time; conceptually, we refine intermediation as ongoing design-after-design that sustains security literacy in later life; and pragmatically, we offer four actionable design moves—trusted accompaniment, slow-paced security onboarding, peer guardians, and low-threshold escalation—that PD teams can adapt in community programs.

Structure of the Paper. Section 2 reviews related work on older adults’ technology use, digital security, and mediation. Section 3 describes the Smartphone Café model and research context and methods of the study. Section 4 presents findings on barriers, mediation strategies, and security practices. Section 5 discusses implications for PD theory and practice. Section 7 concludes with directions for future work.

2 Literature Review

2.1 Technology Appropriation and Security Literacy

Technology appropriation refers to how individuals integrate digital tools into their daily routines and social practices [44, 54]. For older adults, appropriation is shaped by complex cognitive, physical, and experiential barriers [17, 27, 32, 37, 41]. A core component of this appropriation process is digital literacy, which extends beyond

operational skills to include critical understanding of information, privacy, and risk management. Kärnä et al. [29] conceptualize it as a multi-level construct encompassing societal, institutional, and individual dimensions.

Security-related digital literacy remains a significant research gap, leaving many older adults vulnerable to phishing scams, fraud, and misinformation [13, 14, 22, 31, 40, 46, 49, 53]. Research suggests that cybersecurity concerns discourage online participation, reinforcing digital exclusion. Morrison et al. [40] highlight that “the stress associated with cybersecurity could lead some older adults to disengage from security behaviors altogether.” Similar to technology appropriation barriers, physical and cognitive limitations also affect security learning. Reduced vision and dexterity make complex authentication methods difficult, while cognitive decline hinders understanding security settings and fraud detection. According to Sánchez [49], “Participants expressed their idea of manipulation as an abstract concept, hinting at an unknown ‘someone’ behind it. [. . .] They feel technology goes too fast (‘things have changed overnight’) and they find it hard to catch up.” This highlights how cognitive barriers impact older adults’ understanding of online security risks, making them more vulnerable to digital threats. Beyond cognitive and physical limitations, social isolation and lack of peer support further limit opportunities for learning about online security best practices. Vrhovec et al. [59] emphasize that cybersecurity competence among older adults is influenced by their familiarity with technology and access to support networks, but other factors, such as stress, financial situation, and perceived responsibility for online safety, also play a role. To enhance digital security literacy, cybersecurity education must be integrated into broader digital literacy programs. Research suggests that hands-on training, peer-led learning environments, and community-based workshops (such as Smartphone Cafés) are particularly effective in increasing confidence, security awareness, and digital inclusion [18, 62]. These initiatives not only teach technical skills but also reduce technology anxiety and promote self-efficacy, empowering older adults to engage with digital environments safely and independently.

2.2 Social Support and Mediation in Participatory Design

Technology appropriation among older adults is strongly influenced by social context and self-efficacy. Social support, ranging from emotional encouragement to instrumental assistance, plays a crucial role in enabling digital engagement [3, 27, 28]. Research highlights the importance of trusted networks in helping users overcome initial fear and uncertainty, particularly through inter-generational or peer learning formats [10, 30, 61]. Perceived self-efficacy, or belief in one’s capacity to use technology, is likewise central. Programs that build confidence through tailored guidance and repeated exposure tend to foster more sustainable digital engagement [33]. These dimensions suggest that digital inclusion is not simply a matter of access or skills training, but of building supportive, participatory ecosystems.

Recent work in PD highlights the significance of mediators or intermediaries, actors who support marginalized participants in

navigating socio-technical systems and engaging with design processes. These figures often operate in informal or under-recognized capacities, helping to translate between user needs, cultural context, and system constraints [9, 52, 55].

Recent work by Teli et al. [55] further conceptualizes intermediary roles through a tripartite lens: material (providing infrastructure and resources), social (building trust and continuity), and discursive (translating needs and framing inclusion). These roles are particularly important when working with older adults, migrants, and other groups whose participation in design processes may be constrained by socio-technical barriers.

In distributed PD contexts, the role of intermediaries becomes especially crucial. Shidende et al. [52] introduce the concept of "implementation mediators", actors who facilitate the ongoing negotiation between local users, global developers, and institutional constraints during and after initial system deployment. This resonates with our findings, where Smartphone Café facilitators act as boundary-spanning figures who not only teach technical skills but foster relational trust and collective confidence.

2.3 Research Gap

Despite growing recognition of participatory infrastructures in PD, little is known about how informal, community-based settings, such as the Smartphone Café, support long-term digital inclusion through embedded mediation. Much of the existing literature in HCI and PD still focuses on short-term design interventions, often overlooking how participation is enacted over time and across everyday contexts.

This study addresses this gap by examining how older adults and mediators co-create learning environments that support not just technical skill development, but shared understandings of digital risk, trust, and agency. It extends PD scholarship by focusing on sustained co-learning relationships, mediation, and the value of hybrid spaces where participation is ongoing, embedded, and relational.

To avoid conceptual drift noted in prior work, we distinguish social infrastructure (everyday care/support arrangements), participatory infrastructure (ongoing scaffolds for co-participation), and participatory ecosystem (wider multi-stakeholder networks that sustain programs over time). These distinctions frame our analysis of mediation and appropriation in the Smartphone Café. We adopt them as analytic lenses in the findings and implications that follow (see Appendix B for a side-by-side comparison).

3 Methodology and Context

3.1 Context & Background

The primary research site was the so-called Smartphone Café program, established and funded in 2019 by the municipality of a small city in western Germany. The program provides a supportive and informal environment for older adults to explore digital tools such as smartphones, laptops, and tablets, under the supervision of trained mediators. Sessions are inclusive, free, and adaptable to diverse participant needs. Sessions, lasting 2.5 hours, are held on the first Wednesday of each month. Participants enjoy refreshments in a relaxed setting. Attendance ranges from 5 to 10 individuals, mostly aged 50 and above. Over time, a regular cohort of attendees has

developed, though new participants continue to join via word-of-mouth, local advertisements, and a dedicated WhatsApp group that shares session details and updates.

In addition to these monthly group sessions, the program includes office hours (individual consultations) and themed afternoons (topic-based group discussions), each held every six weeks. These sessions address the same target group and also take approx. 2.5 hours.

Field engagement began in June 2023, with researchers attending sessions to build trust and rapport. Over time, informal conversations revealed insights into participants' security practices, including reading Google Play reviews and printing suspicious emails as a self-devised risk mitigation strategy.

The two university researchers and two trained student assistants supported the sessions, in effect acting as mediators or educators at times. They received pre-session training and participated in evaluations before and after each meeting. These evaluations focused on facilitation strengths, participant preferences, group dynamics, and improvements in pedagogical strategy.

3.2 Methods

We employed an ethnographic approach to explore the challenges and barriers faced by older adults and mediators in community-based digital literacy programs, with a particular focus on cybersecurity and digital privacy. It was carried out in a "Smartphone Café" in a rural town in western Germany. The research consisted of ethnographic fieldwork, employing observations and semi-structured interviews to examine participants' experiences with digital technology and cybersecurity, the work of the mediators in the café and the interactions between participants.

The study also investigated mediators' strategies and pedagogical frameworks. Grounded in PD and community-based research principles, the approach emphasized embedded interaction in trusted, low-threshold environments such as municipal Smartphone Cafés and social support organizations. This methodology was chosen for its suitability in working with older and migrant populations who face linguistic, cognitive, and technological challenges.

3.3 Participants and Data Collection

Data collection consisted of observations and interviews with participants and mediators, as described above, during the Smartphone Café settings.

Researchers conducted unstructured observations during sessions, focusing on participant–mediator interactions, group dynamics, and common challenges. The flexible, unstructured format allowed the team to capture both expected and unexpected insights. Observational notes were taken during sessions and later reviewed and supplemented with audio recordings (where consent was granted) to ensure accuracy and depth.

Pre- and post-session evaluations were conducted by the research team, which consisted of two primary researchers and two trained student assistants. These evaluations aimed to reflect on group dynamics, facilitation strengths, and areas for improvement in the pedagogical process.

Semi-structured interviews were conducted face-to-face during Smartphone Café sessions, either individually or in small groups.

Table 1: Overview of participants in the Smartphone Café sessions (excluding mediators).

ID	Gender	Age	Occupation	Primary concerns / topics (security-relevant in <i>italics</i>)	Confidence
P1	Female	69	Retired baker	WhatsApp Status; Amazon ordering; <i>suspicious links/emails/calls</i> ; keeps passwords in notebook; prefers laptop	Improved
P2	Male	60+	NA	Maps/navigation while walking; address input	Improved
P3	Male	NA	NA	Facebook and email issues; <i>strong passwords</i> ; <i>phishing awareness</i>	Improved
P4	Female	NA	NA	WhatsApp group video calls; sharing photos via WhatsApp	Medium
P5a	NA	NA	NA	Online US visa application; <i>avoid fraudulent sites</i> ; <i>verify official websites</i>	Improved
P5b	NA	NA	NA	Online US visa application; <i>avoid fraudulent sites</i> ; <i>secure data handling</i>	Improved
P6	Male	NA	NA	SIM usage; unknown calls; <i>foreign messages</i> ; <i>scam SMS</i>	Medium
P7	Female	70	Retired butcher	Basic phone use; step-by-step guidance	Low
P8	Female	70	Retired nurse	Online health portals; <i>privacy awareness</i>	Medium
P9	Female	70	Retired homemaker	Family communication; <i>avoiding scams</i>	Medium
P10	Female	68	Retired lawyer	<i>Sensitivity to scams</i> ; cautious with personal data	High (cautious)
P11	Female	65	Former librarian	E-books; device settings confusion	Low–Medium
P12	Male	72	Retired factory worker	Calls; photos; avoids digital banking	Low

Notes. NA = not available (participant did not disclose). Confidence reflects self-reported comfort during/after sessions (collapsed to Low / Low–Medium / Medium / High or “Improved” from pre/post reflections). All entries are anonymized; potentially identifying details were generalized. Participation and data use followed informed consent procedures.

Written consent was obtained for individual interviews, while group discussions proceeded with verbal consent. Interview guidelines began with demographic questions and general inquiries about device usage and purposes.

Table 1 gives an overview over the people we interacted with in the context of the Smartphone Café, apart from the mediators.

Empirical corpus (overview). The Smartphone Café study reported in this paper is based on a focused subset of the broader CrossComITS [1] research program; importantly, the data analysed here were collected *prior to the paper submission on 13 June 2025*. We draw on six months of ethnographic fieldwork in two small towns in western Germany. During this period, we attended the regular monthly Smartphone Café sessions (2.5 hours each; 6 sessions) and additionally observed the program’s complementary formats, including office hours (individual consultations) and themed afternoons (topic-based group discussions; held approximately every six weeks; approximately 4 sessions each). Our corpus combines (i) unstructured observations documented in fieldnotes, supplemented by audio recordings where consent was granted, (ii) pre- and post-session evaluation notes produced by the research team, and (iii) semi-structured interviews conducted during sessions, individually or in small groups with older adult participants (Table 1) and the two mediators. All materials were anonymised and imported into MAXQDA [58] for analysis.

3.4 Data Analysis

Data were analyzed using MAXQDA [58] (see Appendix B for the codebook). The analysis employed a thematic coding approach, integrating both predefined categories and emergent codes to capture a comprehensive understanding of participants’ experiences and mediators’ strategies in digital literacy education. Thematic analysis followed Braun and Clarke’s six-phase framework [6], and was adapted for HCI contexts using methodological guidance from Blandford et al. [5].

Data sources included observational notes, interview transcripts, and audio recordings and were triangulated to ensure consistency and validate key findings. Data analysis was conducted by the authors, who collaboratively engaged in coding and interpretation of the qualitative material, ensuring consistency and reflexivity throughout the process.

Reporting conventions. To preserve anonymity and because quotes from group settings were not systematically linked to unique identifiers, we attribute quotations generically (e.g., *Participant*, *interview*; *Participant*, *café session*; *Mediator*) and report qualitative summaries rather than exact counts.

3.5 Research Team and Positionality

The research team consisted of university-based investigators and trained graduate assistants. Both researchers and local facilitators acted as mediators during the sessions, providing technical assistance, emotional reassurance, and personalized guidance to participants. These mediators supported participants in navigating

smartphone use, security concerns, and digital tasks in real time. While some mediators were affiliated with municipal programs, researchers themselves also stepped into this mediating role, allowing them to build trust, support learning, and observe challenges as they emerged. However, at times researchers acted as co-mediators (e.g., guiding a checklist). We mitigated role-bias by alternating roles with municipal staff, memoing decisions, and triangulating fieldnotes/interviews. We note the dual-role as a strength for insight and a limitation for inference.

3.6 Ethical Considerations

This study followed strict ethical guidelines and received approval from the affiliated institution. Written consent was obtained from participants prior to individual interviews, and verbal consent was used for group discussions. Prior to each session it was announced that researchers were present and would collect data in observations, and participants had the option to ask questions, voice concerns or opt out of taking part of the study (they would still be able to participate in the session, but no notes would be taken about them). Interview participants were given the option to decline audio recording. Anonymity and confidentiality were maintained throughout to ensure participants felt safe sharing their experiences.

4 Findings

Dataset context. We participated in a series of Smartphone Café sessions, complemented by group interviews, to explore how older adult participants engage with and adapt to digital technologies. During these sessions, participants were provided with hands-on assistance and guidance in using their smartphones and other digital devices. The group interviews allowed for a deeper exploration of their experiences, challenges, and attitudes towards technology. These interactions provided valuable insights into the practical issues faced by older adults when using digital tools and revealed the emotional and social factors that influence their technology appropriation. The following findings detail the key themes that emerged from these sessions and interviews.

Across the corpus, we repeatedly observed peer help during sessions, a clear preference for in-person learning among interviewees, and widespread reliance on paper-based password practices.

4.1 The Smartphone Café: Where Skills Meet Social Inclusion

Amongst our participants, familiarity with smartphones was minimal, although some had more expertise than others in using a smartphone. Familiarity ranged from using it solely to check the weather forecast, horoscopes and phone calls, to using WhatsApp and YouTube.

Nevertheless, or perhaps because of this, participants consistently highlighted the value of learning new skills and the importance of social interaction in the Smartphone Café. One participant noted that attending these sessions was not only about acquiring new knowledge but also about meeting new people and enjoying the social aspect of the gatherings. People chit-chat, help each other, and also engage in joint "extracurricular activities" such as cooking events, organized by mediators. The sentiment *"And yes, you always*

learn something new. Something. And meet nice people. That's important to me too," reflects the dual benefit of these sessions, enhancing technical skills while fostering a sense of community.

The motivation for attending the Smartphone Café was consistently tied to a desire to learn and maintain social connections. One participant articulated this dual motivation succinctly: *"Yes, my motivation. I'd like to learn too. Social contacts and everything"*. This reflects the café's role not just as a learning environment but also as a social hub for participants.

Participants who had attended the café regularly since its inception expressed satisfaction with the format and the specific topics covered. One participant emphasized their regular attendance and the usefulness of special topic sessions, especially for newcomers, which facilitated both learning and social integration: *"No, no, nothing bothers me. I've actually been here from the beginning. And I come more regularly and also for certain topics... I met them here"*.

Learning outcomes varied among participants, with many expressing that they had gained practical skills that improved their ability to use their smartphones. For instance, one participant mentioned learning how to better handle photography apps and how to combine applications to reduce clutter on their phone. The phrase *"At some point you always learn something new"* was repeated, underscoring the continuous learning process facilitated by the sessions. Another participant reflected, *"I just put more things on it [my smartphone]. At first I couldn't do so much with it,"* referring to their growing ability to install and explore apps over time. This suggests increased confidence and curiosity as a result of the sessions.

The preference for in-person interaction over online alternatives was also evident. When asked if they would prefer the Smartphone Café to be offered online via platforms like Zoom, one participant strongly preferred the in-person format, citing the ease of verbal expression and hands-on interaction: *"No, I prefer to come here"* and emphasized the ease of communication that in-person sessions offer: *"It's just easier to express and ask and do things verbally than to communicate via something like that"*. This preference reflects a broader sentiment among the older adults that physical presence allows for more direct and effective communication, as well as easier clarification and assistance when compared to virtual alternatives.

The impact of the Smartphone Café on participants' lives extended beyond technical skills to include enhanced social interactions. When asked how the café had improved their life, one participant highlighted the importance of social contacts and the enjoyment derived from games: *"Yes, it improves social contacts and games and everything"*. Another participant summed up their motivation as wanting *"to learn, social contacts and everything,"* suggesting that the social aspect is just as important as the educational component. This suggests that the café not only facilitates digital literacy but also contributes to the overall well-being of participants through social engagement.

Participants also discussed their experiences with digital communication tools like WhatsApp and Facebook Messenger. While these tools were generally appreciated for their ability to connect people across long distances, there were concerns about quality and reliability. For instance, a participant mentioned issues with hands-free systems in cars, where incoming calls were often difficult to understand. Similarly, another participant shared frustrations with poor internet connections affecting the quality of video calls.

These experiences underscore the need for better infrastructure and user-friendly technology that can accommodate the specific needs of older adults participants, such as clear audio during calls and reliable connectivity.

Regarding digital literacy, participants generally acknowledged improvements in their technical skills since attending the café. One participant affirmed this, stating, “Yes, *that’s right*,” when asked if they felt their technical skills had improved. However, there was a clear preference for in-person sessions over online alternatives, with one participant noting, “No. *I’m not that far*,” indicating that the face-to-face interaction was more aligned with their comfort level and learning style.

Overall, these findings suggest that the Smartphone Café is not only an effective means of improving digital literacy among older adults participants but also serves as a crucial social space where participants can connect with others, learn collaboratively, and build confidence in using technology. The preference for in-person sessions further emphasizes the need for accessible, community-based learning environments that cater to the unique needs of older adults.

4.2 Privacy Concerns and Password Practices: The Case for Targeted Security Literacy

Participants demonstrated a range of practices and attitudes toward password management, with some relying on their smartphones’ automatic saving features while others were less aware of the potential risks involved. One participant admitted, “Yes, *everything is saved*,” when asked how they manage passwords, implying reliance on the device’s built-in capabilities. This method, while convenient, raises concerns about what might happen if the device is lost or damaged. The participant had not considered the implications of forgetting passwords, reflecting a gap in preparedness: “Yes, *I haven’t thought about that yet*” This lack of foresight underscores the need for increased awareness and education about the importance of secure and accessible password management.

Some relying on traditional methods such as writing them down and hiding them for security. One participant explained, “*You have to write them down. Keep them secret*”. However, this method posed challenges when access to the password was needed away from home, as the participant would have to return home to retrieve the paper: “*So, I have the paper. Then I look at the paper*,” highlighting the inconvenience and potential risk associated with this method.

During the discussion, when a participant casually mentioned their password aloud, it became clear that there is a lack of understanding about the importance of keeping such information confidential. One of the researchers gently responded, “*You shouldn’t actually say your password*”. The participant’s response, followed by laughter, highlights on the one hand the trust participants might have placed in the environment, which enabled them to feel comfortable enough to share their password openly, but on the other hand perhaps also a problematic lack of concern about the risks of security breaches.

Security-related experiences were also explored, with one participant revealing limited experience in this area. When asked about bad experiences with viruses or fraud, the participant responded negatively and admitted to having little experience with password

management: “*I have to be honest: How do I save the password? Because I haven’t done much so far*”. This highlights a gap in knowledge and experience related to digital security, underscoring the need for targeted education in this area.

The discussion also highlighted the potential for biometric security features as a more secure alternative for older adults participants. An interviewer mentioned that biometric options, such as facial recognition, could provide enhanced security: “*There are functions on the phone that...recognize your eye and only then can someone access it*”. This suggestion points to the value of integrating user-friendly, advanced security features to protect sensitive information without relying on memory or written records.

Password security and management emerged as a significant concern for participants. Many found it difficult to remember passwords and resorted to writing them down on paper and storing them in a folder. While the “*trick with the sentence*” (using a memorable phrase to create a strong password) was known, it was not widely used. This indicates a gap between awareness and practice, suggesting that more practical and accessible methods for secure password management should be introduced and reinforced.

Concerns about data security and privacy were prevalent among the participants. One participant expressed uncertainty and fear regarding online data security, stating, “Yes, *data security, I’m not so sure about that sometimes. I’m scared. Because you hear too much*”. This fear was compounded by a lack of comprehensive understanding, leading to cautious behavior online. For example, the participant refrained from returning calls from unknown numbers and shared an anecdote about a warning from a friend regarding scam phone calls: “*There are wrong addresses here with WhatsApp again and then he wrote not to press the blue button*”. This reflects a reliance on community advice and face-to-face warnings to navigate potential online threats.

Privacy concerns were also noted, particularly in situations where participants were unable to view someone else’s profile picture, leading to uncertainty about whether they had been blocked or if the person simply did not have a profile picture.

Security and Trust issues were recurring concern among participants, particularly regarding online banking and password management. One participant, a woman with medium digital competence, recounted her child’s unfortunate experience with online banking fraud, which reinforced her decision to avoid online banking altogether. This highlights a broader issue of trust in digital financial transactions among older adults.

Participants also voiced a preference for personal contact when seeking help with security issues or device problems. The suggestion to allow questions to be asked without requiring registration reflects a desire to minimize barriers to seeking help and to maintain privacy. Moreover, the concept of a secure, device-integrated password management feature, potentially implemented as a pop-up box, was suggested as a way to address concerns about remembering passwords.

Another significant area of concern related to privacy, was particularly in using location services with Google Maps. One participant was initially hesitant to enable location access due to privacy concerns. However, after discussing the safety and utility of the feature, she enabled it and subsequently learned to navigate using both walking and driving modes. This case underscores the importance

of addressing privacy concerns directly and providing clear explanations to build trust and confidence in using digital tools.

Furthermore, participants expressed a strong preference for supporting local businesses over online shopping. As one participant explained, *“They don’t do online shopping because they want to support local stores; they only buy things they cannot get locally from trusted shops”*. This reflects a community-oriented mindset and a potential resistance to the impersonal nature of online transactions, which might be less appealing to this demographic.

Past negative experiences with online shopping significantly influenced participants’ attitudes. For instance, one couple decided to quit using eBay after a bad experience, stating, *“They have quit eBay after bad experiences (they sent a package, and it was broken...)”*. Such experiences contribute to a reluctance to engage in online activities, highlighting the importance of building trust and ensuring positive experiences in digital environments for older adults.

Overall, the findings suggest that while older adults participants exhibit a range of digital behaviors, from minimal use to active engagement with specific functions, they often rely on basic security practices and community support to navigate digital threats. The reliance on written passwords, limited digital activity, and concerns about data security highlight the need for targeted educational interventions to enhance digital literacy, particularly in the areas of online safety and secure password management. These interventions could help build confidence and ensure that older adults can safely and effectively navigate the digital landscape.

4.3 Mediators and Facilitation Strategies

Mediators played a central role in creating an engaging and inclusive learning environment. Mediator 1 (M1), a municipal employee involved since 2022, helped integrate Themennachmittag sessions with the Smartphone Café. Mediator 2 (M2), a freelance digital educator, brought specialized knowledge in digital literacy training. Across sessions, mediators also deliberately cultivated a relaxed, social atmosphere that made it easier for participants to voice concerns and uncertainties: *“We do chit chats to soften the atmosphere, so they feel relaxed and talk about their questions and problems.”*

To build trust and foster community, mediators and researchers also engaged in informal social activities such as cooking events and casual games (e.g., ping pong) at the café. These activities helped strengthen relationships and make participants feel comfortable expressing their challenges and questions.

The café’s physical setup supported small-group interaction: tables for 3–4 participants, shared Wi-Fi, and hands-on troubleshooting using participants’ own devices. Free internet access was provided, and mediators assisted participants in connecting their personal devices—including smartphones, tablets, or laptops. Instruction was personalized based on each participant’s skill level. Topics ranged from basic tasks (e.g., sending messages or managing photo galleries) to more complex issues such as password recovery or identifying phishing attempts. Within this participatory infrastructure, we repeatedly observed four mediating practices that shaped how security learning happened in situ: *trusted accompaniment, slow-paced security onboarding, peer guardianship, and discursive reframing*.

Trusted accompaniment (staying-with through risky or uncertain moments). Participants often framed help-seeking as relational: instead of “figuring it out alone,” they preferred a known and trusted person to stay with them while navigating unfamiliar steps (e.g., recovery flows, updates, unknown messages). This was visible in the way participants described what they would do when a phone was lost or compromised. One participant described immediately turning to the mediator as a first response: *“I call M1 first, I would have to report to the police? I don’t know.”* Such accounts show that “security” was not only a technical matter but also a question of *who* could be relied on for calm, non-judgmental assistance. Mediators also built this trust through informal social activities (e.g., games, cooking events), which lowered the threshold for asking “small” questions that often had security implications.



Figure 1: Trusted mediators facilitating group interaction and digital engagement

This emphasis on personal contact and trust was also echoed by mediators, who described their role as accompanying participants through uncertainty rather than simply delivering instructions.

Slow-paced security onboarding (step-by-step, repeatable routines). A recurring challenge was that security-relevant actions (password practices, privacy settings, updates) were experienced as fragile: participants feared “breaking something” or losing access. Mediators responded with slow, step-by-step guidance and encouraged participants to externalise memory demands (notes, routines). Participants’ password practices illustrate this tension. Some relied on device auto-saving (*“Yes, everything is saved”*), while others used paper-based strategies: *“You have to write them down. Keep them secret.”* In one moment, when a participant casually spoke their password aloud, a researcher intervened gently—*“You shouldn’t actually say your password”*—highlighting how the café’s supportive atmosphere could simultaneously enable openness and require careful boundary-setting. Across these interactions, “onboarding” meant more than explanation: it involved pacing, repetition, and making security steps feel doable. For instance, when a participant

struggled with the “forgot password” function, the mediator demonstrated the process step by step, then guided the participant through the task on their own device.

Peer guardianship (participants helping participants to triage risk). Beyond mediator support, participants also relied on peers and everyday networks to interpret suspicious messages, calls, or links. This sometimes happened in-session (regulars assisting newcomers) and sometimes through external peer warnings that were brought into the café as “evidence” for discussion. For instance, one participant described receiving a warning from a friend about suspicious WhatsApp prompts: “There are wrong addresses here with WhatsApp again and then he wrote not to press the blue button.” Such peer-to-peer advisories functioned as lightweight “guardianship,” where participants circulated simple rules (“don’t press the blue button,” “don’t answer unknown numbers”) to reduce risk. The café amplified this by turning individual alerts into shared learning moments: mediators invited others to share their strategies and normalised checking with the group before acting.

We also observed peer learning emerging organically: after receiving help with password recovery, the same participant later assisted others facing similar issues, illustrating how support practices circulated among regular attendees and newcomers.

Discursive reframing (making abstract threats understandable). Mediators frequently translated security concepts into relatable metaphors to reduce anxiety and support confident action. When discussing suspicious links and phishing, a mediator reframed the threat in everyday terms: “Think of this as a stranger knocking on your door without proper ID.” These analogies helped participants evaluate risk without needing technical language, and they legitimised “pausing” before clicking as a sensible, everyday precaution. Discursive reframing also supported privacy-related decisions, such as enabling location services for navigation: when participants hesitated due to privacy concerns, mediators explained trade-offs in concrete terms and guided them through settings changes. In effect, language worked as a security tool: it made threats discussable and transformed fear into actionable judgement.

At the same time, mediators noted tensions arising from the wide range of participant skill levels: some attendees experienced the sessions as repetitive, while others felt overwhelmed. These dynamics highlight the need for more customizable materials and structured opportunities for peer-to-peer support within community-based learning infrastructures. Across these practices, M2 emphasized the importance of design and facilitation approaches that alleviate anxiety, encourage calm exploration, and foster curiosity rather than expecting immediate results. He advocated offering alternatives and allowing participants to solve problems at their own pace, supported by trust, assurance, and collaborative problem-solving.

5 Discussion

This section synthesizes the study’s findings with existing literature and explores broader implications for older adults’ digital engagement, particularly through the lens of Smartphone Café sessions. In addressing our research questions, we examine (RQ1) how mediation within participatory infrastructures shapes older adults’ appropriation of smartphone practices for digital security literacy;

(RQ2) the cognitive, physical, and experiential barriers that arise in everyday use; (RQ3) the mediating practices—material, social, and discursive—that enable confidence and safer behavior; and (RQ4) how these practices carry forward beyond the Smartphone Café through peer help and emerging routines.

Our findings confirm and expand prior research on digital inclusion, social learning, and cybersecurity literacy by emphasizing the central role of mediated, relational learning in enabling older adults to engage with digital technologies in safe, meaningful ways.

5.1 Challenges in Technology Appropriation and Security Concerns

Older adults face persistent challenges in appropriating digital technologies, consistent with earlier research [7, 36]. Participants in the Smartphone Café sessions struggled with basic and advanced digital tasks, such as managing photos, using maps, and handling updates in applications like WhatsApp. These barriers stem not only from cognitive and physical limitations but also from limited early exposure to digital tools [7, 12, 47]. Participants expressed frustration with rapid interface changes and unnecessary updates, which they felt increased confusion rather than offering value. This highlights the need for digital systems designed with simplicity and predictability for aging users.

However, the challenges extend beyond usability. Our findings show that emotional responses to digital security are particularly strong. Participants expressed anxiety, fear, and hesitation when faced with online services, citing concerns over scams, fraud, and opaque security protocols [12, 15, 41, 56, 57]. These concerns often lead to avoidance behavior, where individuals choose not to engage at all with certain digital services, particularly those involving sensitive information [19, 33].

Our findings confirm earlier results that security-related challenges among older adults fall into three key categories. First, cognitive and physical barriers, such as declining memory, slower processing speed, and reduced dexterity, impact their ability to manage authentication methods and secure passwords effectively. We saw this in how people struggled to remember passwords. Second, psychological and social barriers, including fear of scams, phishing attacks, and financial fraud, contribute to avoidance behaviors and a lack of confidence in navigating security settings. Finally, usability and systemic barriers, such as opaque security settings and complex authentication processes, create additional obstacles that may expose users to digital risks [14, 39]. These issues are compounded when users opt for convenience over safety, using weak passwords or bypassing warnings, thereby increasing exposure to digital risk [43, 64].

Yet, as our study shows, security is not only a technological problem; it is inherently social. Older adults often learn to manage digital risks not by reading instructions or navigating interfaces alone, but through relational, trust-based encounters with others. At the Smartphone Café, participants received emotional reassurance, peer validation, and real-time guidance, an environment in which fears were acknowledged and learning was shared. This aligns with prior work on “intermediated use” [9] and Akama’s notion of “relational participation” [2], where the presence of mediators transforms uncertainty into agency. For example, a participant who

had long avoided updating WhatsApp due to fears of breaking something shared, after a guided session: “Now that I see how it works, it’s actually easier than I thought.” In another instance, a casual peer conversation about password safety led one attendee to begin using a physical notebook for safe storage. These moments reflect ‘situated learning’, not just instruction, but co-presence, repetition, and trust.

Therefore, community support plays a crucial role in helping older adults navigate digital security challenges [51, 62]. Participants in the Smartphone Café sessions benefited from guided learning approaches, where mediators provided both technical and emotional support. This aligns with previous research emphasizing that social learning and trust-based support systems enhance digital appropriation among older adults [27, 56]. Social support serves multiple functions by reducing technology-related anxiety, providing hands-on assistance with password management and fraud detection, and fostering peer learning through shared experiences.

Security literacy, in this context, emerges from social infrastructures like the Smartphone Café, not merely from interfaces or settings. Mediators played a crucial role not only in providing information, but also in holding space for confidence-building, clarification, and trial-and-error learning. Designing effective security interventions for older adults therefore requires more than accessible technology; it demands systems that acknowledge and support these ‘social modes of learning and care’. Step-by-step training on security best practices, delivered in an accessible and structured format, can help older adults feel more confident in protecting themselves online [45]. Future programs should formalize this insight: integrating peer mentorship, slow-paced explanation, and emotional reassurance into structured digital literacy training. Community-driven, mediator-supported models hold great promise for enabling sustained digital engagement and resilient, self-aware security practices. These patterns clarify **RQ2**: barriers are not solely cognitive or physical but also *experiential*, shaped by prior negative encounters and ambient risk talk. They also speak to **RQ1**, where appropriation of security-relevant routines depends on the presence of mediators and peers who can slow the pace, externalize steps, and normalize rehearsal.

5.2 Mediation and Support Strategies

Community support plays a crucial role in helping older adults navigate digital security challenges. Participants benefited from guided learning approaches in the Smartphone Café sessions, where mediators provided both technical and emotional support. This aligns with previous research emphasizing that social learning and trust-based support systems enhance digital appropriation among older adults [11, 27, 35, 51, 56, 62].

Social support in these sessions functioned in multiple ways. Emotional reassurance was a significant factor, as participants expressed a reduction in technology-related anxiety when engaging in a safe and supportive learning environment. Practical assistance also played a key role, particularly in helping older adults manage passwords, detect fraudulent activities, and navigate complex application interfaces. Additionally, peer learning emerged as an important aspect, as participants shared their experiences and

problem-solving strategies, reinforcing each other’s self-efficacy and confidence in digital security practices.

To answer **RQ3** (Mediating practices), across sessions, four practices repeatedly underpinned safer behavior: (1) *Trusted accompaniment*—a known mediator or peer stays with the participant through the whole task; (2) *Slow-paced onboarding*—see-once/do-once sequences with printed or on-device mini-scripts; (3) *Peer guardianship*—regulars informally triage suspicious messages and updates at the table; (4) *Discursive reframing*—risky actions metaphorized (e.g., “locking the front door”) to reduce anxiety and legitimize action. For example, one participant initially struggled with a WhatsApp security update but, after receiving step-by-step guidance, noted that it was easier than expected. This instance highlights how direct support can demystify security features and encourage continued digital engagement. Similarly, discussions on password management encouraged participants to consider safer digital habits, demonstrating that interactive and peer-supported learning environments effectively enhance cybersecurity competence.

These findings reinforce the importance of mentorship programs and peer-assisted learning as strategies to improve digital security awareness among older adults. Future interventions should incorporate structured training programs that combine technical guidance with ongoing emotional and social support, ensuring that older adults not only gain the necessary skills but also develop confidence in their ability to navigate digital environments securely. These practices persisted beyond sessions (**RQ4**), with participants reporting use at home and relaying them to neighbors and friends.

5.3 The Smartphone Café Initiative

The Smartphone Café initiative served as a critical setting for exploring these interactions. This program showed to provide a supportive and informal environment where older adults can come together to learn and experiment with digital tools under the guidance of mediators. The café sessions were designed to be inclusive and accommodating, addressing the diverse needs and skill levels of participants.

The practical approach of the sessions helped to demystify technology and to build confidence among participants. Müller and Struzek [42] describe “appropriation cafés”, a similar format, as key to the PD process in the Cognitive Village project, where older adults explored and learned to use digital technologies in a supportive environment. Their sessions helped participants build digital competence and engage in discussions on future technological developments, fostering motivation and meaningful participation in the design process [42].

A key feature of the Smartphone Café is its focus on personalized support. Mediators offer one-on-one assistance, answering questions and providing step-by-step guidance tailored to the individual’s pace and understanding. This personalized attention helps alleviate the anxiety and frustration that often accompany learning new technologies, making the experience more positive and empowering. According to Hornung et al. [24] “Informal coffee-and-cake-settings helped provide a cozy and casual environment in order to prevent stress and limit frustration in their learning experiences”. Our study shows that these findings also extend to matters of security and privacy, and their respective peculiarities.

Moreover, the café fosters a community atmosphere where participants can share their experiences and learn from each other. This peer support is invaluable, as it provides social reinforcement and encourages continued engagement with technology. The collaborative environment also allows for the exchange of tips and strategies, further enhancing the learning experience. According to Liu et al. [33] “that promoting social engagement can motivate people to engage in more complex interactions, mobilize cognitive abilities, and help maintain good mental health”.

By observing and engaging with participants in these sessions, this study leverages valuable insights into the digital habits, preferences, and concerns with regard to security and privacy of older adults. The interactions recorded during the café sessions reveal common challenges, such as difficulties with app navigation, security concerns, and preferences for certain types of digital content and services. These observations are crucial for understanding the real-world experiences of older adults and identifying effective strategies to support their digital literacy journey.

Framed as a *participatory infrastructure*, the Café coordinates material scripts, social roles, and discursive framings that sustain learning between sessions rather than delivering one-off training. This suggests design implications: invest in stable mediator roles, printable micro-scripts for risky actions, and lightweight peer roles that can be enacted outside formal programming.

6 Implications for PD and Educational Interventions

Our findings show that addressing digital exclusion among older adults is about more than simply applying user-centered design (UCD) principles. While UCD is widely accepted in the design industry, many digital products still fail to meet the needs of older adults, not necessarily due to a lack of awareness, but because their needs are often de-prioritized or considered too complex to integrate into mainstream development. Rapid development cycles, youth-centric markets, and assumptions about digital familiarity can overshadow the specific usability and accessibility needs of older populations. As studies show, older adults often struggle to use digital tools compared to younger generations [23, 36, 48].

Rather than focusing solely on design principles, it is essential to prioritize the lived experiences and adaptive strategies of older adults. In our study, participants often faced challenges with newer features, even when they had mastered basic digital tasks such as messaging or photo browsing. These findings align with those of Bertolazzi et al. [4], who point to the cumulative effect of sensory limitations, cognitive change, and low self-confidence in learning as barriers to sustained digital appropriation. Our study emphasizes that these challenges are not abstract problems, but are experienced daily through frustration, uncertainty, and fear.

Muller et al. [41] emphasize a sensitive and user-centric approach to technology appropriation, ensuring that interventions are practically beneficial and emotionally supportive, enhancing older adults' ability to navigate the digital world confidently. Czaja et al. [12] discuss tailored educational programs that are tailored to meet the learning needs of older adults. Interventions such as repetition, simplified instructions, and hands-on practice can improve older adult learners' retention and ease of use. In light of cognitive challenges

such as memory decline and slower information processing, these pedagogical techniques are vital for older adults to acquire and maintain digital skills effectively.

Yap et al. [63] note that “older adults technology appropriation offers several benefits, such as improving quality of life [34], enriching lives through gameplay [60], enhancing social interactivity and cognitive skills, and decrease depression [47], and promoting a more independent and healthier lifestyle [8]”. Pihlainen et al. [48] argue that “engagement in education and learning in various fields—digital or not—can have profoundly positive effects on older adults' quality of life, their outlook on life, and images of ageing [26], particularly in the context of digital technologies”. Mitzner et al. [36] emphasize that technology adoption in later life is not a “one-off decision” but an extended process that develops as needs arise and use becomes routinized. Müller and Struzek [42] state that “Older people want to understand what they can use technology for, so it has to be meaningful for them to embark on the learning journey”.

However, these design and educational interventions are essential for bridging the digital divide, a gap that not only limits access to technological benefits but also exacerbates social isolation. By ensuring that digital tools are accessible and that older adults are equipped with the skills and confidence to use them, we can promote greater digital inclusion. According to Liu et al. [33] “many cases of social isolation and rejection are frequently found among the older adults population; these were commonly correlated with the weakening of the person's mental health, which can cause mental disorders such as depression”.

Our study highlights the broader social implications of digital exclusion among the older adults. The preference for analog methods and the avoidance of online activities not only limit access to the benefits of digital technologies but also contribute to social isolation. As more services move online, the risk of marginalizing those who are digitally excluded increases. Therefore, fostering digital inclusion through accessible design, supportive learning environments, and trust-building initiatives is essential. Such efforts can enhance the quality of life for older adults by enabling them to participate fully in the digital society, stay connected with loved ones, and access essential services. In their study, Guo [20] noted that “respondents were also frustrated by the lack of human contact with like-minded peers to learn together, which is consistent with findings on the greater social isolation experienced by older adults as a group”. Mohan and Lyons [38] state that “the wellbeing of older people is supported where family members assist in learning and problem solving, where tolerance and patience boost confidence, though frustration across generations in technological ability is also recognized”. Furthermore, concerns about media storage, privacy, and managing multiple accounts were prevalent, indicating a need for practical tips and reliable tools. According to Hornung [24] “There are many reasons for the difficulties with IT appropriation and acceptance among the older adults, ranging from privacy and security to concerns about self-presentation”.

Our work shows that it is possible and successful to replicate experiences of learning with peers and assistance and problem solving that others have reported, with regards to the use of digital media and cyber security. This is especially relevant as it is vital to address and overcome fears related to security risks that Hornung

reported [24] and which we also detailed above, as they inhibit digital inclusion and participation of older adults.

A key feature of the Smartphone Café model is the role of mediators, volunteers, municipal staff, and younger participants who support older adults during learning. These mediators were not only technical guides, but also emotional and social facilitators. They translated abstract instructions into meaningful action, offered reassurance during failure, and paced the sessions in ways that matched participants' comfort. Their presence made the difference between confusion and understanding, hesitation and confidence.

7 Conclusion

This study emphasizes the importance of multifaceted support systems for older adults in adopting digital technologies, particularly in addressing security and privacy concerns. The findings highlight the necessity of designing digital environments that accommodate the cognitive and emotional needs of older adults while alleviating security-related anxieties. Ensuring that digital tools and educational programs are accessible and tailored to this demographic is essential for fostering greater digital inclusion and online safety.

From a theoretical perspective, this research contributes to the fields of HCI and PD by demonstrating how older adults' digital security concerns intersect with social learning, and the crucial role of mediators and mediation in creating suitable learning environments. While previous studies have focused on general digital literacy barriers, this study expands digital inclusion models by emphasizing the role of real-time mediation and peer learning, specifically in addressing security anxieties. The findings reinforce the idea that social support networks play a crucial role in helping older adults build confidence in digital security practices.

On a practical level, this study underscores the need for intuitive security designs and tailored cybersecurity education for older adults. Developing user-friendly password management tools, simplifying authentication processes, and incorporating peer-led learning environments can significantly enhance digital confidence and engagement. Designing secure digital platforms that prioritize clarity and ease of use will help reduce the cognitive burden associated with navigating security settings.

Additionally, community-based programs such as Smartphone Cafés provide a scalable and effective model for embedding cybersecurity literacy into informal learning settings. These initiatives not only offer technical guidance but also create supportive environments where older adults can learn collaboratively and reinforce their digital skills through social interactions. Future research should further investigate the long-term impact of such interventions on older adults' digital autonomy and overall well-being.

By addressing the key challenges older adults face in digital security and literacy, this study contributes to the broader discourse on digital inclusion, security education, and community-based learning interventions. Creating educational and technological solutions that cater to the specific needs of older adults will promote safer, more inclusive participation in the digital world, ultimately enhancing their ability to navigate digital spaces with confidence and security.

However, this study is based on a single municipal Smartphone Café initiative in two small towns in western Germany, where a

stable in-person program, trusted mediators, and recurring participation shaped how security learning became possible. The practices we describe—trusted accompaniment, slow-paced onboarding, peer guardianship, and discursive reframing—are likely to travel to other community-based digital support settings, but their form may differ where funding is unstable, staffing is volunteer-only, participation is more transient, or language and migration histories create additional mediation needs. Rather than claiming direct generalisability, we offer these practices as transferable design sensitivities that can guide adaptation and evaluation in other local infrastructures.

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A Appendices

B Concept Distinctions

Table A1: Distinguishing social infrastructure, participatory infrastructure, and participatory ecosystem.

Term	Scope	Core actors	Temporality	Governance	Example in our case
Social infrastructure	Supports everyday interaction and care	Community orgs, volunteers, venues	Ongoing	Informal/formal local rules	Café venue, regulars, municipal staff
Participatory infrastructure	Arrangements enabling <i>ongoing co-participation</i> in shaping practices/services	Participants, mediators, designers	Program cycles; iterative	Shared norms; light scaffolds	Session scripts, peer roles, co-authored checklists
Participatory ecosystem	Network linking programs, policies, services, markets	NGOs, city services, vendors, clinics	Multi-year	Multi-stakeholder	Café ↔ municipal IT helpdesk ↔ telecom shop

C MAXQDA Codes

▼ ● 📁 Code System	141
● 📁 Digital Tools Usage	2
● 📁 game	8
● 📁 Device Accessibility	4
● 📁 Experience with Digital Communication	10
● 📁 Concerns About Digital Interaction	2
● 📁 User Guidance & Informative Solutions	4
● 📁 Technical Issues & difficulties	26
● 📁 Nervousness and Anxiety with Technology	1
● 📁 Digital Literacy and Competence	1
▼ ● 📁 Motivation and Benefits of visit the smartphone café	9
● 📁 Quality of Life Improvement	1
● 📁 Learning and Social Interaction	5
● 📁 Social Interaction	2
▼ ● 📁 Communication Preferences	4
● 📁 Expert Chat	1
● 📁 Personal Contact & In-Person vs. Online	3
▼ ● 📁 User Preferences	8
● 📁 Digital Hesitation	2
● 📁 Preferences for Analog vs. Digital	3
▼ ● 📁 Terminology and Language	2
● 📁 Menu Labels	3
● 📁 Clear Language	5
▼ ● 📁 Content and Features	1
● 📁 Video Integration	1
● 📁 Interactive Exercises	1
▼ ● 📁 Security Concerns	15
● 📁 Online Security and Fraud Experiences	1
● 📁 App Reliability	2
● 📁 Password Management	7
▼ ● 📁 App Usability	4
● 📁 Interface Clarity	2
● 📁 Learning Curve	1
● 📁 Sets	0